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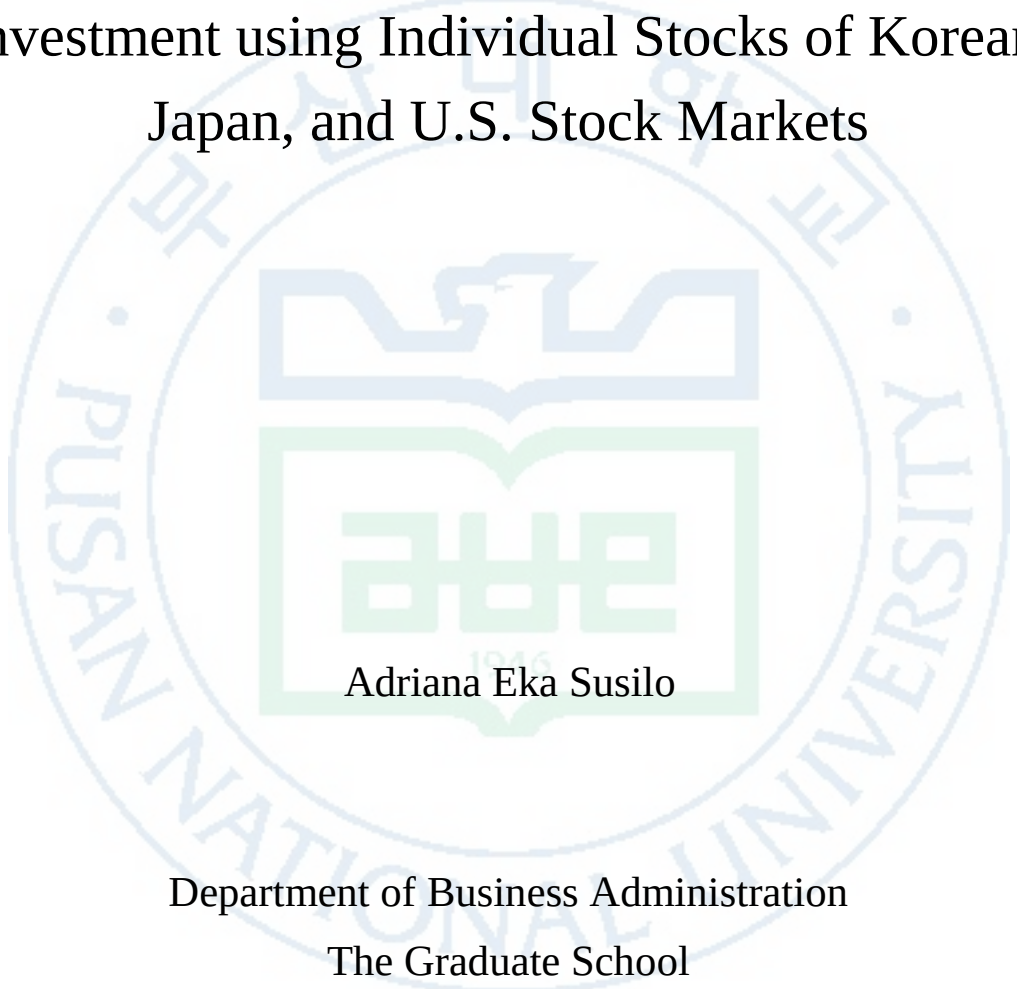
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Thesis for the degree of Master of Arts in Business Administration

Profitability of Moving Average in Portfolio
Investment using Individual Stocks of Korean,
Japan, and U.S. Stock Markets

The background of the page features a large, faint watermark of the Pusan National University logo. The logo is circular, with the university's name in English, "PUSAN NATIONAL UNIVERSITY", around the perimeter. In the center is a stylized emblem consisting of a blue crown-like shape above a green square containing a white geometric pattern. The year "1946" is also visible within the emblem.

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August 2018

Profitability of Moving Average in Portfolio Investment
using Individual Stocks of Korean, Japan, and U.S.
Stock Markets
by Adriana Eka Susilo

A thesis submitted to the graduate school of
Pusan National University in partial fulfillment of
the requirements for the degree of Master of Arts in Business Administration
under the direction of Cheoljun Eom

The thesis for the degree of Master of Arts in Business Administration
by Adriana Eka Susilo
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Profitability of Moving Average in Portfolio Investment using Individual Stocks in Korean, U.S., and Japan Stocks Markets

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Abstract

In this thesis, I present supporting evidence about profitability of moving average timing strategy in investment return when applied in decile volatility portfolios in Korean, U.S., and Japan Stocks markets. The moving average timing strategy generates higher return or outperform the respective buy-and hold strategy. This strategy also still generates significantly strong result even after considering transaction costs. This positive return of MAPs have little risk exposure of CAPM, Fama-French 3 factors model, and also momentum factors model. Moreover, the results are robust even when using alternative moving average length and value-weighted size decile portfolios. The abnormal return of MAPs also cannot be explained by market timing factor even though they generate a significant successful market timing strategy.

1 Introduction

1.1 Motivation and Background

Technical analysis has been widely applied in financial market for a long time by practitioners to analyze financial data and make trading decisions. Technical analysis uses past data to predict future market movement. Even though practitioner mainly use technical analysis, it is less popular than fundamental analysis in the academic area. Researcher is mainly focus on the fundamental analysis in modern quantitative finance.

Numerous empirical studies find strong evidence for the profitability of technical analysis. Brock, Lakonishok, and LeBaron (1992) find evidence that simple technical indicator, such as moving average, have significant ability to predict stock return. Lo, Mamaysky, and Wang (2000) also find that technical analysis has predictive ability and can add value to the investment process. Neely et al. (2014) show that technical indicator is statistically significant in-sample and out-of-sample predictive power as macroeconomic variables in forecasting stock market.

Almost all of the previous research has weigh more on the time-series data and individual stocks return in technical analysis calculation. Recently, Han et al. (2013) find evidence that technical analysis has cross-sectional profitability in US stock market. Han et al. (2013) stated that the more uncertain the information is or the more volatile the stock is, the more profitable the technical analysis will be.

This thesis is similar in spirit to Han et al. (2013). However, there are several differences stand out. First, in calculating MA timing strategy I use moving average for individual stock instead of portfolios. Second, I use excess return instead of average return of the portfolio. Because excess return can describe better than the average return. Third, I use Korean, U.S. and Japan stock market data as a basis for the calculation. Many researchers also start to investigate and focus on Korean and Japan market because globalization makes Korean and Japan market rapidly grow. The different characteristic of Korean and Japan with U.S. stock markets make many researchers interested in them.

Korean market is known to be different with U.S. stock market so the cross-sectional profitability of technical analysis in Korean market might be different with the cross-sectional profitability of technical analysis in U.S. market. Korean stock market is known to have dominant individual investors while the U.S. stock market is known to have dominant institutional investors (Jaewoo Kang, 2017). Korean market also has low volatility effect which high volatility portfolio will generate lower return compared to the low volatility portfolio (Kho and Kim, 2014). Yamada and Nagawatari (2011) find evidence that Japan also generates low volatility effect because of the over-estimation from Japan retail investor.

1.2 Objectives

The objective of this thesis is to empirically present the evidence supporting that investment return using moving average timing strategy may achieve higher performance than buy-and-hold return, when using portfolio constructed based on previous volatility (annual standard deviation), even though considering transaction cost. This profitability or abnormal return is still cannot be explained well by the popular CAPM, Fama-and-French 3 factors models, and Carhart four-factors models.

The main contribution of this thesis is in the following: (i) show support evidence about the profitability of moving average timing strategy in Korean, U.S., and Japan stocks market; (ii) investigate the profitability or abnormal return by the popular CAPM, Fama-and-French 3 factors models, and Carhart four-factors models; and (iii) investigate the profitability or abnormal return by market timing ability.

1.3 Organization of This Thesis

Chapter 2 Literature Review presents the technical analysis and moving average timing strategy. Chapter 3 presents the evidence of cross-sectional profitability of MA Timing strategy in Korean, U.S. and Japan Market by the significantly positive return of

MAP and capital asset pricing models (CAPM), Fama-and French 3 factors, and four-factor Carhart (1997) model cannot fully explain about the abnormal return of the MAPs. The robustness of MA timing strategy profitability will be discussed in Chapter 4. Chapter 5 presents that market timing model explain the abnormal return of MAPs but do not fully explain them. Finally, this thesis concludes in Chapter 6 Conclusions.



2 Literature Review

2.1 Technical Analysis

Technical analysis uses past data and other statistic to predict future market movement. Technical analysis is already used in the investment analysis by practitioners from 1800s but it is still concluded useless by academic area (Brock, Lakonishok, and LeBaron (1992). However, numerous empirical studies found strong evidence for the profitability of technical analysis. Brock, Lakonishok, and LeBaron (1992) find evidence that simple technical indicator, such as moving average, have significant ability to predict stock return. Although just using simple technical tools like moving average and trading break, they provide strong support for technical analysis. Lo, Mamaysky, and Wang (2000) also find that technical analysis has predictive ability and can add value to the investment process. Neely et al. (2014) show that technical indicator is statistically significant in-sample and out-of-sample predictive power as macroeconomic variables in forecasting stock market

The previous research has weigh more on the time-series data and individual stocks return in technical analysis calculation. Recently, Han et al. (2013) find evidence that technical analysis has cross-sectional profitability in U.S. stock market. They focus on the portfolios sorted by volatility than on the individual stocks. Han et al. (2013) stated that the more uncertain the information is or the more volatile the stock is, the more profitable the technical analysis will be. When the market is uncertain, investors tend to focus more on the technical indicator because the fundamental analysis will differ more from the actual price.

2.2 Moving Average Timing Strategy

The following exposition of the MA strategy follows closely the presentation in Han et al. (2013) but with a modification in the calculation of MA Portfolio. Han et al (2013) use portfolio price when calculating the MA portfolio but I use individual stocks in calculating MA timing strategy. After some set of stocks are sorted into decile portfolios,

MA timing strategy is used for individual stocks in each portfolio. Let R_{jt} ($j = 1, \dots, 10$) be the return on individual stocks, and P_{jt} ($j = 1, \dots, 10$) be the corresponding individual stock prices. The MA of individual stock at day t with lag L is define as:

$$A_{jt,L} = \frac{P_{jt-(L-1)} + P_{jt-(L-2)} + \dots + P_{jt-1} + P_{jt}}{L} \quad (1)$$

which is the average price of the past L days including day t . I use an MA of length $L = 10$ days as baseline case same with Han et al. (2013). The MA investment strategy is to compare the closing price on each trading day t . If the last closing price P_{jt-1} is above the MA price $A_{jt-1,L}$, the strategy is to invest in that stock j for the next trading day t ; otherwise if the last closing price P_{jt-1} is below the MA price $A_{jt-1,L}$, the strategy is to invest in the risk free asset.

Mathematically, the returns on the MA timing strategy are:

$$\tilde{R}_{jt,L} = \begin{cases} R_{jt}, & \text{if } P_{jt-1} > A_{jt-1,L}; \\ r_{ft}, & \text{otherwise,} \end{cases} \quad (2)$$

Where R_{jt} is the return on the j th individual stocks on day t , and r_{ft} is the return on the risk-free asset. Return of the portfolio and then constructed with the equal-weighted based calculation. Zero-cost arbitrage portfolio, MAP, is constructed as the difference between the return of the MA strategy for portfolio j at time t and the return of buy-and-hold strategy portfolio j at time t , as follows:

$$MAP_{jt,L} = \tilde{R}_{jt,L} - R_{jt}, \quad j = 1, \dots, N. \quad (3)$$

A MAP is a portfolio that takes a long position in the MA timing portfolio and a short position in the underlying portfolio to determine the relative performance of the MA strategy against buy-and-hold strategy. The significant abnormal return of the MAPs can be interpreted as evidence of the profitability of the MA timing strategy over the buy-and-hold strategy of the underlying portfolio.

For the calculation, I use daily equal-weighted returns of sets of 10 volatility decile portfolios sorted by volatility. I use Korean, U.S., and Japan stocks market for calculation and daily data of all stocks are readily available from fnGuide. The decile portfolios for Korean stocks are constructed based on the KOSPI and KOSDAQ stocks sorted into 10

groups by their annual standard deviations estimated using the daily returns within the prior year. The portfolios are rebalanced each year at the end of the previous year. The sample period starts in July 1, 2000 and ends in June 30, 2017.



3 Profitability of MA Timing Strategy

In this section, I present the summary statistic of the volatility decile portfolio (panel A), MA timing portfolio (panel B), and the correspondent MAPs (panel C), the alphas of MAPs from CAPM and Fama-and-French 3 factors model, and alphas from Carhart four-factors model. Finally, I discuss about the reasons for the profitability of the MA timing strategy.

3.1 Summary Statistics

Table 1 reports the characteristics of the return on the underlying portfolios by buy-and-hold strategy, 10-days MA timing portfolios, and the excess return (MAP) of the MA timing strategy over the buying and holding of the underlying portfolio in Korean stock market. The average returns are all annualized and t-statistics value are in parentheses. The average excess returns of Korean stock market in Panel A are a decreasing function of the volatility decile portfolios. The last row in the table provides the difference between the highest and the lowest deciles portfolios. The difference between the highest and lowest decile is -0.15 (t-statistic -2.45). The negative relation between volatility and the return in Korean stock market is because of the low volatility effect in the Korean Stock market and it is consistent with Kho and Kim (2014) paper. But, the MA timing portfolio return in Panel B have a positive relation with volatility as the return has increased across the decile portfolios ranging with 0.518 (t-statistic 15.28) difference of the highest and lowest decile portfolio. The different relation of the MA timing portfolio could be because of the significant decrease of standard deviation in each decile portfolio. The standard deviation in MA timing strategy is much smaller compared with the underlying portfolios under buy-and-hold strategy and the difference of the standard deviation in panel B is not as much as the difference of the standard deviation in panel A. For example, in the lowest decile portfolio, standard deviation is 2.72 vs 0.44 and for the highest decile is 5.88 vs 0.94. MA strategy outperforms buy-and-hold strategy as the average excess returns of the MA strategy are higher than the average excess return of the underlying portfolios which

makes the return of the MAP portfolios (panel C) are all positive. The different trend in panel A and panel B makes the difference of

Rank	Panel A Volatility Decile Portfolios		Panel B MA(10) Timing Portfolios		Panel C MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	0.075 (2.08)	2.721	0.396 (10.96)	0.440	0.322 (10.51)	2.651
2	0.045 (1.06)	3.166	0.502 (12.17)	0.504	0.457 (14.34)	3.100
3	0.071 (1.71)	3.214	0.540 (11.57)	0.576	0.469 (13.53)	3.135
4	0.045 (1.13)	3.711	0.597 (12.67)	0.619	0.552 (12.06)	3.641
5	0.005 (0.17)	4.030	0.648 (12.56)	0.677	0.642 (11.63)	3.955
6	-0.001 (-0.02)	4.276	0.651 (11.93)	0.704	0.652 (12.21)	4.178
7	0.013 (0.23)	4.867	0.726 (11.73)	0.779	0.713 (10.34)	4.745
8	-0.022 (-0.44)	4.900	0.753 (11.92)	0.806	0.775 (13.70)	4.798
9	-0.035 (-0.51)	5.678	0.847 (12.76)	0.883	0.881 (11.96)	5.569
High	-0.084 (-1.05)	5.882	0.914 (15.65)	0.938	0.998 (13.85)	5.750
High-Low	-0.159 (-2.45)		0.518 (15.28)		0.677 (11.78)	

Table 1. Summary Statistics of Korean Stock Market

the highest and lowest decile portfolio become higher around 0.68. The standard deviation in panel C is reported slightly smaller than in the panel A.

Table 2 reports the characteristics of the return on the underlying portfolios by buy-and-hold strategy, 10-days MA timing portfolios, and the excess return (MAP) of the MA timing strategy over the buying and holding of the underlying portfolio in U.S. stock market. These results are consistent with Han (2013) paper. The average returns in panel A are an increasing function of the decile, with the difference between highest and lowest

decile portfolio is 0.13. The characteristic of panel B and C in U.S. stock market are the same with Korean market. Returns in panel B and C are also an increasing function of the decile portfolio with difference between highest and lowest decile portfolios are 0.54 and 0.40.

Rank	Panel A Volatility Decile Portfolios		Panel B MA(10) Timing Portfolios		Panel C MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	-0.093 (-0.64)	2.3640	0.285 (5.12)	0.4317	0.3781 (1.97)	9.7353
2	0.020 (0.76)	2.6459	0.336 (6.50)	0.3940	0.3160 (4.44)	2.5814
3	0.024 (1.17)	2.5716	0.363 (9.31)	0.3658	0.3398 (6.86)	2.5136
4	0.034 (1.14)	3.2024	0.416 (14.33)	0.4060	0.3825 (8.09)	3.1388
5	0.034 (0.93)	2.9946	0.471 (12.84)	0.4359	0.4364 (7.89)	2.9070
6	0.052 (1.77)	3.1650	0.521 (12.98)	0.4849	0.4682 (8.92)	3.0756
7	0.030 (0.96)	3.5568	0.577 (13.21)	0.5482	0.5469 (9.73)	3.4554
8	0.062 (1.33)	4.6530	0.637 (11.82)	0.6195	0.5749 (7.25)	4.5587
9	0.017 (0.31)	4.6187	0.725 (11.57)	0.7438	0.7086 (7.97)	4.4939
High	0.045 (0.86)	4.7318	0.826 (10.32)	0.8080	0.7806 (8.71)	4.6323
High-Low	0.1379 (1.03)		0.5404 (11.17)		0.4025 (2.94)	

Table 2. Summary Statistics of U.S. Stocks Market

The standard deviation in panel B is much smaller compared with the corresponding decile in panel A. Panel C returns are also all positive and significant indicating the better performance of MA timing portfolios.

Table 3 reports the characteristics of the return on the underlying portfolios by buy-and-hold strategy, 10-days MA timing portfolios, and the excess return (MAP) of the MA timing strategy over the buying and holding of the underlying portfolio in Japan stock

market. Unlike the result in Korean and U.S. stock market that has relation with volatility, Japan records no trend in decile portfolios under buy-and-hold strategy. The average stock

Rank	Panel A		Panel B		Panel C	
	Volatility Decile Portfolios		MA(10) Timing Portfolios		MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	-0.271 (-1.64)	9.667	0.316 (14.07)	0.552	0.5875 (3.57)	9.579
2	-0.045 (-0.83)	6.507	0.389 (12.05)	0.658	0.4334 (8.26)	6.408
3	-0.358 (-1.33)	14.716	0.419 (11.85)	0.733	0.7774 (2.58)	14.625
4	-0.170 (-0.88)	10.415	0.442 (11.87)	0.764	0.6117 (3.31)	10.334
5	0.025 (0.46)	5.762	0.473 (13.01)	0.828	0.4475 (8.50)	5.654
6	0.008 (0.18)	6.097	0.510 (11.80)	0.887	0.5020 (7.75)	5.987
7	0.112 (1.81)	6.364	0.564 (12.03)	1.071	0.4525 (6.23)	6.247
8	-0.088 (-1.21)	7.189	0.532 (12.77)	0.911	0.6192 (7.97)	7.066
9	-0.019 (-0.20)	7.297	0.571 (13.42)	1.015	0.5898 (5.24)	7.151
High	-0.337 (-1.14)	13.176	0.608 (13.00)	1.014	0.9458 (3.03)	13.062
High-Low	-0.0661 (-0.28)		0.2923 (7.31)		0.3584 (1.45)	

Table 3. The Summary Statistics of Japan Stocks Market

returns and standard deviation in panel A have no trend as the return vary across decile portfolios. But when we just look the highest and the lowest portfolios, the result is consistent with Yamada and Nagawatari (2011) which stated that Japan also generates low volatility effect. In panel B, the returns of the MA portfolios are an increasing function of the decile portfolios but the difference between the highest and lowest portfolios is smaller than Korean and U.S. stock market. Japan also generates successful MA timing strategy as the returns of MAP in panel C all positive and significant.

This summary statistic shows that MA timing strategy performs very well compared to the buy-and-hold strategy. A different result in panel A of Korean and U.S. market is caused by investors situation differences between these two countries. Korean stock market is dominant with individual or retail investors, while institutional investors take a dominant role in U.S. stock market. Stocks that have higher volatility are tend to have more retail investor that makes the relation between risk and return become different or called anomaly (Jaewoo Jang, (2017), Kho and Kim (2014)).

3.2 Transaction Cost

The profitability of moving average timing strategy above may be affected when we included transaction costs in calculation. In this section, I will show the evidence that the profitability of the MA timing strategy is still strong even though considering transaction cost.

Following Han et al. (2013) paper, the return on the MA timing strategy in the presence of transaction cost τ per trade are:

$$\tilde{R}_{jt,L} = \begin{cases} R_{jt}, & \text{if } P_{jt-1} > A_{jt-1,L} \text{ and } P_{jt-2} > A_{jt-2,L}; \\ R_{jt} - \tau, & \text{if } P_{jt-1} > A_{jt-1,L} \text{ and } P_{jt-2} < A_{jt-2,L}; \\ r_{ft}, & \text{if } P_{jt-1} < A_{jt-1,L} \text{ and } P_{jt-2} < A_{jt-2,L}; \\ r_{ft} - \tau, & \text{if } P_{jt-1} < A_{jt-1,L} \text{ and } P_{jt-2} > A_{jt-2,L}. \end{cases} \quad (4)$$

I use the same sample period and same lag-length with the previous test. I assume transaction cost for Korean market is 0.35% roundtrip following Kho and Kim (2014) paper, U.S. market is 0.25% one-way following Lynch and Balduzzi (2000) and Han et.al (2013) paper, and 0.3% roundtrip for Japan market following Hu (1998) paper.

Table 4,5, and 6 report the average annual excess return and standard deviation of the decile portfolio by buy-and-hold strategy, 10-days MA timing strategy, and the difference between them (MAP) in Korean, U.S. and Japan stocks market. The returns are all annualized and the t-statistic values are in parentheses. All of stock market generate the

same pattern with previous result and transaction cost does not affect much the return of MA timing strategy.

	Panel A		Panel B		Panel C	
Rank	Volatility Decile Portfolios		MA(10) Timing Portfolios		MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	0.075 (2.08)	2.7208	0.332 (9.07)	0.4702	0.257 (8.39)	2.6621
2	0.045 (1.06)	3.1656	0.436 (10.49)	0.5336	0.391 (12.19)	3.1083
3	0.071 (1.71)	3.2145	0.474 (10.09)	0.6067	0.403 (11.56)	3.1445
4	0.045 (1.13)	3.7112	0.530 (11.22)	0.6461	0.485 (10.64)	3.6543
5	0.005 (0.17)	4.0305	0.582 (11.19)	0.7008	0.577 (10.33)	3.9679
6	-0.001 (-0.02)	4.2757	0.585 (10.67)	0.7315	0.586 (10.90)	4.1847
7	0.013 (0.23)	4.8668	0.659 (10.62)	0.8036	0.647 (9.37)	4.7592
8	-0.022 (-0.44)	4.8997	0.685 (10.89)	0.8338	0.708 (12.51)	4.8146
9	-0.035 (-0.51)	5.6783	0.780 (11.75)	0.9049	0.815 (11.11)	5.5773
High	-0.084 (-1.05)	5.8824	0.848 (14.51)	0.9570	0.933 (12.91)	5.7582
High-Low	-0.159 (-2.45)		0.517 (15.21)		0.676 (11.74)	

Table 4. Summary Statistics of Korean Stock Market including transaction cost

In Korean stock market, the differences before and after deducting transaction cost are almost stable among decile portfolios. Transaction cost makes return of MAP decrease around 7-20% from previous result. Although the return of MAP decreased, it still generates the significantly positive result. This result suggests that MA timing strategy

still outperforms buy-and-hold strategy even if we consider transaction cost in the calculation in Korean stock market.

Rank	Panel A		Panel B		Panel C	
	Volatility Decile Portfolios		MA(10) Timing Portfolios		MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	-0.093 (-0.64)	2.3640	0.199 (3.83)	0.4902	0.2915 (1.47)	9.7440
2	0.020 (0.76)	2.6459	0.243 (4.99)	0.4365	0.2227 (3.10)	2.5801
3	0.024 (1.17)	2.5716	0.267 (7.62)	0.4026	0.2434 (5.09)	2.5136
4	0.034 (1.14)	3.2024	0.314 (10.53)	0.4370	0.2804 (5.67)	3.1362
5	0.034 (0.93)	2.9946	0.369 (9.76)	0.4670	0.3352 (5.76)	2.9067
6	0.052 (1.77)	3.1650	0.421 (10.27)	0.5115	0.3685 (6.72)	3.0745
7	0.030 (0.96)	3.5568	0.477 (10.68)	0.5758	0.4468 (7.69)	3.4564
8	0.062 (1.33)	4.6530	0.537 (9.78)	0.6402	0.4758 (5.78)	4.5606
9	0.017 (0.31)	4.6187	0.627 (9.82)	0.7622	0.6104 (6.67)	4.4964
High	0.045 (0.86)	4.7318	0.728 (8.96)	0.8154	0.6831 (7.39)	4.6356
High-Low	0.1379 (1.03)		0.5295 (11.52)		0.3916 (3.06)	

Table 5. Summary Statistics of U.S. Stocks Market including transaction cost

Same with Korean stock market, transaction cost does not affect much the return of MAP. Transaction cost affects the return of MAP around 12-30% in U.S. Stock market. Despite the decrease of MAP return, MA timing strategy still outperforms buy-and-hold strategy in U.S stock market with considering transaction cost.

Rank	Panel A		Panel B		Panel C	
	Volatility Decile Portfolios		MA(10) Timing Portfolios		MAP	
	Avg Ret	Std Dev	Avg Ret	Std Dev	Avg Ret	Std Dev
Low	-0.271 (-1.64)	9.6673	0.253 (9.81)	0.6387	0.5247 (3.20)	9.5667
2	-0.045 (-0.83)	6.5071	0.325 (10.22)	0.7440	0.3699 (3.72)	6.3840
3	-0.358 (-1.33)	14.7156	0.356 (10.28)	0.8111	0.7135 (2.42)	14.6164
4	-0.170 (-0.88)	10.4153	0.377 (11.21)	0.8253	0.5467 (3.23)	10.3354
5	0.025 (0.46)	5.7620	0.410 (13.00)	0.8856	0.3845 (7.87)	5.6536
6	0.008 (0.18)	6.0967	0.446 (11.36)	0.9488	0.4388 (7.29)	5.9961
7	0.112 (1.81)	6.3644	0.502 (11.11)	1.1037	0.3902 (5.62)	6.2402
8	-0.088 (-1.21)	7.1886	0.468 (12.26)	0.9754	0.5553 (7.50)	7.0631
9	-0.019 (-0.20)	7.2966	0.508 (12.31)	1.0795	0.5265 (5.08)	7.1646
High	-0.337 (-1.14)	13.1756	0.546 (11.42)	1.0650	0.8830 (2.76)	13.0651
High-Low	-0.0661 (-0.28)		0.2923 (7.32)		0.3584 (1.45)	

Table 6. Summary Statistics of Japan Stocks Market including transaction cost

Same with Korean and U.S. stock markets, transaction costs also do not affect the return of MAP in Japan stock market. MAP returns still generate significantly positive values. This result concludes that MA timing strategy still generates significant higher performance than buy-and-hold return even though considering transaction cost in Korean, U.S., and Japan stock markets.

3.3 Risk-Adjusted Return or Alpha

The asset pricing models I consider in this section are CAPM:

$$MAP_{jt,L} = \alpha_j + \beta_{j,MKT} r_{MKT,t} + \epsilon_{jt}, \quad j = 1, \dots, 10 \quad (5)$$

and Fama-and-French (1993) 3-factor model:

$$MAP_{jt,L} = \alpha_j + \beta_{j,MKT} r_{MKT,t} + \beta_{j,SMB} r_{SMB,t} + \beta_{j,HML} r_{HML,t} + \epsilon_{jt}, \quad j = 1, \dots, 10 \quad (6)$$

where $r_{MKT,t}$ is the daily market excess return, $r_{SMB,t}$ is the daily return of the SMB factor, and $r_{HML,t}$ is the daily return of the HML factor. Table 7, 8, and 9 reports the parameter estimates of the regression for each factors model for Korean, U.S., also Japan stocks markets. The alpha values are annualized and the t-statistics are in parentheses. The risk-adjusted alphas are all positive and significant for both CAPM and FF-3 factor models for all markets. The positive and significant value of alphas show that the profitability of MA timing strategy and both CAPM and FF-3 factor models cannot fully explain the abnormal return of MAPs. When comparing CAPM alphas and Fama-and-French 3 factors model, Fama-and-French 3 factors alphas are smaller than the risk-adjusted CAPM alpha suggest that Fama-and-French 3 factors model is slightly better in explaining the return of MAP. The risk-adjusted alphas have the same trend and higher than MAP return for all of the market because of the negative value of market betas (Han, 2013).

The loading factor of market has a significant role in explaining the abnormal return of MAPs but the loading factor of SMB and HML do not have significant role in explaining the abnormal return of MAPs for both Korean and Japan markets. The market betas of all of the stock markets generate significant result while, the betas for SMB and HML generate insignificant result for all of the sock markets.

Rank	CAPM		Fama-French				R ²
	α_j	$\beta_{j,MKT}$	α_j	$\beta_{j,MKT}$	$\beta_{j,SMB}$	$\beta_{j,HML}$	
Low	0.3328	-0.0417	0.3222	-0.0342	0.0176	0.0364	11.29
	(10.72)	(-2.07)	(11.16)	(-1.73)	(1.28)	(1.89)	
2	0.4770	-0.0124	0.4611	-0.0033	0.0136	0.0424	8.12
	(15.07)	(-1.08)	(16.65)	(-0.31)	(1.19)	(1.64)	
3	0.4945	-0.0732	0.4859	-0.0619	0.0133	0.0392	10.88
	(15.45)	(-3.00)	(15.89)	(-2.57)	(0.93)	(2.11)	
4	0.5774	-0.1145	0.5523	-0.0848	0.0516	0.0870	21.04
	(14.95)	(-3.14)	(15.13)	(-3.19)	(2.07)	(2.54)	
5	0.6721	-0.1158	0.6577	-0.0943	0.0229	0.0692	18.69
	(12.31)	(-2.48)	(12.33)	(-2.46)	(1.18)	(1.55)	
6	0.6840	-0.1055	0.6674	-0.0789	0.0238	0.0939	13.16
	(13.38)	(-2.94)	(12.61)	(-2.47)	(1.22)	(3.46)	
7	0.7571	-0.1152	0.7318	-0.0903	0.0034	0.0828	10.40
	(11.29)	(-2.69)	(10.50)	(-2.33)	(0.16)	(2.05)	
8	0.8144	-0.0845	0.7997	-0.0729	-0.0091	0.0484	13.66
	(15.11)	(-2.10)	(14.82)	(-1.80)	(-0.39)	(1.38)	
9	0.9229	-0.0788	0.8838	-0.0644	-0.0252	0.1088	12.01
	(12.75)	(-1.97)	(11.90)	(-1.76)	(-1.00)	(2.48)	
High	1.0409	-0.0317	0.9985	-0.0084	-0.0300	0.1319	10.72
	(14.89)	(-1.15)	(13.87)	(-0.26)	(-1.42)	(3.20)	
High-Low	0.7081	0.0100	0.6763	0.0258	-0.0476	0.0955	
	(12.79)	(0.29)	(11.54)	(0.73)	(-2.12)	(3.28)	

Table 7. CAPM and Fama-French Alphas for Korean Stocks Markets

Rank	CAPM		Fama-French				R ²
	α_j	$\beta_{j,MKT}$	α_j	$\beta_{j,MKT}$	$\beta_{j,SMB}$	$\beta_{j,HML}$	
Low	0.4011	-0.0024	0.3793	-0.0029	-0.0006	-0.0058	11.01
	(2.12)	(-2.45)	(1.94)	(-2.03)	(-0.63)	(-1.01)	
2	0.3396	-0.0023	0.3362	-0.0023	-0.0003	0.0000	19.83
	(5.25)	(-3.94)	(4.94)	(-4.13)	(-0.45)	(0.10)	
3	0.3709	-0.0028	0.3589	-0.0027	-0.0001	0.0002	29.26
	(8.24)	(-4.54)	(8.06)	(-4.24)	(-0.34)	(0.54)	
4	0.4274	-0.0028	0.4166	-0.0025	-0.0005	-0.0002	22.68
	(9.98)	(-5.15)	(9.51)	(-4.63)	(-1.51)	(-0.47)	
5	0.4758	-0.0039	0.4689	-0.0037	-0.0007	-0.0003	32.56
	(10.62)	(-9.04)	(10.23)	(-9.32)	(-1.92)	(-0.92)	
6	0.5215	-0.0041	0.5163	-0.0038	-0.0011	-0.0004	32.19
	(12.11)	(-8.01)	(11.56)	(-8.26)	(-2.28)	(-0.99)	
7	0.6075	-0.0042	0.6004	-0.0038	-0.0012	0.0000	25.38
	(12.81)	(-6.43)	(12.35)	(-6.69)	(-2.23)	(0.09)	
8	0.6400	-0.0038	0.6376	-0.0035	-0.0007	-0.0005	18.37
	(8.58)	(-4.45)	(8.56)	(-5.03)	(-0.97)	(-1.05)	
9	0.7719	-0.0037	0.7607	-0.0031	-0.0010	0.0002	14.01
	(9.07)	(-3.79)	(8.92)	(-3.95)	(-1.35)	(0.32)	
High	0.8356	-0.0044	0.8291	-0.0040	-0.0006	-0.0001	15.69
	(9.65)	(-5.00)	(9.45)	(-5.63)	(-0.63)	(-0.11)	
High-Low	0.4345	-0.0020	0.4498	-0.0010	0.0001	0.0057	
	(3.13)	(-1.38)	(3.03)	(-0.62)	(0.05)	(1.03)	

Table 8. CAPM and Fama-French Alphas for U.S. Stocks Markets

Rank	CAPM		Fama-French				R ²
	α_j	$\beta_{j,MKT}$	α_j	$\beta_{j,MKT}$	$\beta_{j,SMB}$	$\beta_{j,HML}$	
Low	0.6344	-0.0033	0.6679	-0.0035	-0.0002	-0.0009	13.83
	(3.68)	(-5.87)	(3.18)	(-4.67)	(-0.18)	(-0.24)	
2	0.4714	-0.0037	0.4521	-0.0033	0.0029	0.0018	22.40
	(10.39)	(-3.85)	(10.68)	(-3.88)	(3.22)	(1.55)	
3	0.8502	-0.0048	0.8494	-0.0042	0.0033	0.0036	23.34
	(2.58)	(-4.01)	(2.55)	(-4.33)	(1.55)	(2.10)	
4	0.6564	-0.0043	0.6404	-0.0036	0.0029	0.0006	20.55
	(3.52)	(-4.32)	(3.42)	(-4.20)	(4.25)	(0.71)	
5	0.4964	-0.0049	0.4631	-0.0038	0.0043	0.0007	28.93
	(9.08)	(-3.50)	(9.25)	(-3.35)	(3.14)	(0.67)	
6	0.5468	-0.0048	0.5062	-0.0037	0.0041	0.0011	29.91
	(8.65)	(-3.69)	(8.64)	(-3.71)	(3.95)	(0.70)	
7	0.4997	-0.0050	0.4693	-0.0038	0.0041	0.0015	28.22
	(7.77)	(-3.55)	(8.99)	(-3.73)	(3.65)	(1.00)	
8	0.6694	-0.0047	0.6354	-0.0037	0.0036	0.0020	25.86
	(9.03)	(-4.19)	(8.49)	(-3.75)	(4.06)	(1.59)	
9	0.6479	-0.0052	0.6298	-0.0039	0.0036	0.0002	29.48
	(6.05)	(-3.52)	(6.20)	(-3.10)	(3.77)	(0.15)	
High	1.0154	-0.0053	1.0686	-0.0051	0.0000	-0.0045	26.56
	(2.77)	(-3.68)	(2.31)	(-2.53)	(0.00)	(-0.61)	
High-Low	0.3810	-0.0019	0.4008	-0.0016	0.0003	-0.0036	
	(1.52)	(-2.55)	(1.39)	(-1.47)	(0.12)	(-0.96)	

Table 9. CAPM and Fama-French Alphas for Japan Stocks Markets

3.4. Momentum Factor

Besides CAPM and Fama-and-French 3 factors model, I also consider four-factor Carhart (1997) which is included momentum factor in the regression of Fama-and-French 3 factors model:

$$MAP_{jt,L} = \alpha_j + \beta_{j,MKT}r_{MKT,t} + \beta_{j,SMB}r_{SMB,t} + \beta_{j,HML}r_{HML,t} + \beta_{j,UMD}r_{UMD,t} + \epsilon_{jt}, \quad j = 1, \dots, 10 \quad (7)$$

where $r_{MKT,t}$ is the daily market excess return, $r_{SMB,t}$ is the daily return of the SMB

factor, $r_{HML,t}$ is the daily return of the HML factor, and $r_{UMD,t}$ is the daily return of the UMD factor.

Table 10 reports the regression result of MAP on the Carhart four-factor (1997) for U.S. and Japan markets. Daily data of UMD is available from Kenneth French's data library. I did not include Korean in the calculation because there is no UMD data available for Korean markets. The alpha values are all annualized and the t-statistics are in parentheses. For both U.S. and Japan stock markets, momentum also does not explain the return of MAP. The value of alpha is still positive and significant same with the alpha result from CAPM and Fama-and-French 3 factors model regression. The momentum beta coefficient for both U.S. and Japan market are all negative and mostly insignificant. Even though the explanatory power of 4-factors model is higher than Fama-and-French 3 factors models, we can conclude that 4-factors model or momentum still only explain a small portion of the abnormal return of MAP.

Rank	U.S. Stock Market						Japan Stock Market					
	α_j	$\beta_{j,MKT}$	$\beta_{j,SMB}$	$\beta_{j,HML}$	$\beta_{j,UMD}$	R^2	α_j	$\beta_{j,MKT}$	$\beta_{j,SMB}$	$\beta_{j,HML}$	$\beta_{j,UMD}$	R^2
Low	0.358	-0.0027	-0.0003	-0.0059	-0.0002	12.40	0.624	-0.0031	0.0004	-0.0022	-0.0041	14.37
	(2.10)	(-2.22)	(-0.30)	(-1.05)	(-0.27)		(3.32)	(-3.32)	(0.41)	(-0.68)	(-1.95)	
2	0.331	-0.0023	-0.0003	-0.0001	-0.0003	20.09	0.443	-0.0029	0.0031	0.0011	-0.0014	22.91
	(4.81)	(-3.97)	(-0.42)	(-0.17)	(-1.18)		(11.00)	(-3.30)	(3.14)	(0.98)	(-3.62)	
3	0.347	-0.0026	-0.0001	0.0000	-0.0005	30.45	0.793	-0.0036	0.0037	0.0034	-0.0023	24.18
	(8.28)	(-3.79)	(-0.15)	(0.05)	(-1.74)		(2.37)	(-3.03)	(2.10)	(1.61)	(-1.10)	
4	0.413	-0.0024	-0.0003	-0.0001	-0.0005	23.51	0.643	-0.0029	0.0024	0.0019	0.0013	21.80
	(8.60)	(-4.32)	(-1.08)	(-0.20)	(-1.73)		(2.77)	(-2.90)	(1.92)	(2.13)	(0.39)	
5	0.461	-0.0035	-0.0006	-0.0002	-0.0006	35.34	0.479	-0.0037	0.0042	0.0008	0.0005	29.96
	(9.92)	(-8.81)	(-1.34)	(-0.49)	(-2.21)		(7.29)	(-3.19)	(3.28)	(1.11)	(0.52)	
6	0.512	-0.0037	-0.0009	-0.0004	-0.0006	34.54	0.520	-0.0035	0.0040	0.0006	-0.0006	30.82
	(10.81)	(-7.74)	(-1.83)	(-0.92)	(-1.75)		(7.07)	(-3.65)	(3.90)	(0.49)	(-0.61)	
7	0.593	-0.0036	-0.0010	-0.0002	-0.0009	28.23	0.476	-0.0037	0.0042	0.0013	-0.0007	29.45
	(11.81)	(-6.28)	(-1.75)	(-0.24)	(-1.84)		(6.58)	(-3.82)	(4.24)	(1.31)	(-0.66)	
8	0.629	-0.0033	-0.0005	-0.0009	-0.0005	21.08	0.637	-0.0037	0.0035	0.0026	0.0005	26.78
	(8.52)	(-4.56)	(-0.75)	(-1.46)	(-0.89)		(7.84)	(-3.74)	(3.40)	(2.38)	(0.63)	
9	0.750	-0.0030	-0.0008	-0.0002	-0.0007	16.99	0.636	-0.0037	0.0034	0.0007	0.0007	30.74
	(9.18)	(-3.65)	(-1.08)	(-0.24)	(-1.34)		(5.21)	(-2.95)	(3.31)	(0.54)	(0.64)	
High	0.813	-0.0038	-0.0003	-0.0001	-0.0015	17.63	1.045	-0.0051	0.0004	-0.0043	-0.0014	27.21
	(9.33)	(-5.10)	(-0.32)	(-0.07)	(-2.77)		(2.31)	(-2.24)	(0.14)	(-0.64)	(-1.01)	
High-Low	0.4547	-0.0011	0.0000	0.0058	-0.0013		0.4211	-0.0020	-0.0001	-0.0021	0.0027	
	(3.92)	(-0.75)	(0.01)	(1.07)	(-1.24)		(1.30)	(-1.45)	(-0.03)	(-0.42)	(1.09)	

Table 10. Regression with Momentum Factor

4. Robustness Test

In this section, I examined the robustness of the MAP returns performance. I will show evidence that MA timing strategy performance is still better than buy-and-hold strategy even for the various alternative length of the moving average and also the performance when using the value-weighted size decile portfolios.

4.1. Alternative Lag Length

I use 20- and 50-days MA timing strategy as the alternative lag length in calculating MAP return. Table 11 reports the average excess return and Fama-French (1993) alphas for the MAPs of the various length and stock markets. The results are basically the same with previous result. The MAP returns for all decile portfolios in various length and market are all positive suggest that MA timing strategy performs better than buy-and-hold strategy in various lag-length. The alphas also generate significant and positive results for all decile portfolios. However, the magnitude of MAP returns and alphas decrease as the lag-length increase. This result is consistent with Han (2013) paper.

4.2. Size Decile Portfolios

Table 12 shows evidence that the performance of MA timing strategy is still better than the buy-and-hold strategy even though using value-weighted size decile portfolios for Korean stock markets. The size deciles are ranked using market capitalization of the previous year and rebalanced every year. The sample period of the deciles portfolios is the same with volatility decile portfolios from July 1, 2000 until June 30, 2017. All of the MAP returns and alphas are significantly positive value and the trend for various lag-length is the same with previous result however, the magnitude of abnormal return are decrease compare to the volatility decile portfolios. It shows that profitability of the MA timing strategy is still strong when using value-weighted size decile portfolios.

Rank	US				Korea				Japan			
	MA (20)		MA (50)		MA (20)		MA (50)		MA (20)		MA (50)	
	Avg Ret	α_j	Avg Ret	α_j	Avg Ret	α_j	Avg Ret	α_j	Avg Ret	α_j	Avg Ret	α_j
Low	0.3139	0.3254	0.2300	0.2517	0.2357	0.2462	0.1388	0.1675	0.4854	0.5871	0.4008	0.5795
	(1.83)	(1.89)	(1.43)	(1.41)	(8.55)	(8.82)	(5.13)	(5.38)	(3.05)	(2.87)	(2.57)	(2.59)
2	0.2342	0.2528	0.1484	0.1830	0.3400	0.3603	0.2041	0.2519	0.3313	0.3590	0.2140	0.2829
	(4.17)	(5.09)	(3.56)	(5.17)	(12.94)	(15.39)	(12.21)	(14.44)	(3.58)	(3.47)	(2.27)	(2.25)
3	0.2453	0.2675	0.1416	0.1925	0.3264	0.3572	0.1850	0.2366	0.6751	0.7668	0.5554	0.7525
	(6.85)	(8.68)	(6.13)	(8.95)	(12.05)	(14.32)	(7.16)	(8.22)	(2.36)	(2.33)	(1.94)	(2.05)
4	0.2731	0.3142	0.1731	0.2476	0.3902	0.4110	0.2384	0.2926	0.5088	0.5504	0.3700	0.4847
	(7.47)	(9.48)	(6.54)	(11.06)	(10.90)	(13.56)	(9.07)	(10.31)	(3.02)	(3.10)	(2.19)	(2.36)
5	0.3116	0.3501	0.1854	0.2517	0.4710	0.5049	0.2825	0.3451	0.3141	0.3334	0.1425	0.1816
	(7.33)	(11.28)	(6.48)	(13.53)	(10.84)	(11.64)	(8.55)	(9.22)	(6.88)	(8.59)	(3.95)	(5.43)
6	0.3308	0.3858	0.1961	0.2789	0.4676	0.5062	0.2804	0.3522	0.3757	0.3861	0.2267	0.2698
	(8.33)	(12.72)	(7.84)	(15.70)	(10.44)	(10.91)	(6.99)	(7.50)	(6.64)	(7.64)	(4.52)	(5.13)
7	0.3859	0.4483	0.2320	0.3226	0.4972	0.5378	0.2816	0.3519	0.3263	0.3433	0.1513	0.1888
	(10.02)	(15.07)	(9.41)	(17.46)	(8.37)	(8.46)	(5.69)	(5.98)	(5.10)	(7.21)	(3.49)	(5.82)
8	0.4012	0.4726	0.2281	0.3252	0.5553	0.6028	0.3148	0.3994	0.4806	0.5049	0.2962	0.3727
	(6.75)	(8.49)	(5.39)	(7.05)	(12.99)	(13.55)	(8.47)	(8.71)	(7.33)	(7.92)	(5.37)	(5.91)
9	0.5087	0.5747	0.3188	0.4206	0.6345	0.6632	0.3580	0.4176	0.4259	0.4699	0.2348	0.3170
	(7.74)	(8.99)	(7.34)	(8.16)	(10.15)	(9.61)	(7.91)	(7.70)	(4.42)	(5.27)	(2.85)	(3.67)
High	0.5579	0.6179	0.3498	0.4438	0.7068	0.7360	0.4239	0.5054	0.7777	0.9278	0.5588	0.8014
	(8.48)	(9.78)	(7.64)	(9.35)	(12.10)	(12.05)	(6.89)	(6.90)	(2.54)	(2.27)	(1.95)	(1.89)
High-Low	0.2440	0.2925	0.1198	0.1921	0.4711	0.4898	0.2851	0.3379	0.2923	0.3408	0.1580	0.2219
	(1.79)	(2.05)	(0.80)	(1.19)	(9.37)	(9.09)	(5.43)	(5.21)	(1.21)	(1.15)	(0.69)	(0.70)

Table 11. Alternative Moving Average Length

Rank	MAP(10)		MAP(20)		MAP(50)	
	Avg Ret	α_j	Avg Ret	α_j	Avg Ret	α_j
Large	0.518	0.528	0.355	0.381	0.200	0.252
	(11.21)	(12.43)	(10.13)	(11.91)	(8.40)	(9.87)
2	0.492	0.501	0.339	0.364	0.174	0.217
	(11.70)	(12.49)	(11.25)	(12.31)	(7.85)	(8.95)
3	0.550	0.558	0.392	0.417	0.221	0.268
	(12.03)	(12.17)	(10.62)	(10.90)	(6.70)	(6.88)
4	0.609	0.615	0.440	0.466	0.266	0.323
	(11.31)	(11.09)	(9.88)	(9.70)	(6.50)	(6.54)
5	0.603	0.608	0.433	0.456	0.243	0.295
	(13.74)	(13.57)	(12.06)	(12.01)	(7.50)	(7.54)
6	0.653	0.655	0.475	0.497	0.255	0.305
	(12.23)	(11.42)	(10.51)	(9.91)	(6.25)	(6.00)
7	0.663	0.665	0.475	0.498	0.264	0.317
	(15.72)	(14.98)	(15.08)	(14.43)	(8.94)	(8.52)
8	0.706	0.713	0.508	0.536	0.312	0.371
	(13.29)	(13.13)	(10.42)	(10.11)	(7.37)	(7.28)
9	0.691	0.705	0.493	0.531	0.283	0.351
	(8.55)	(7.98)	(6.54)	(6.07)	(4.13)	(4.04)
Small	0.827	0.826	0.584	0.606	0.352	0.417
	(10.46)	(9.64)	(9.76)	(9.02)	(6.75)	(6.32)
Small-Large	0.308	0.298	0.229	0.225	0.153	0.165
	(5.09)	(4.51)	(4.56)	(4.02)	(3.27)	(2.79)

Table 12. Size Decile Portfolios

5. Drivers of Abnormal Return

In this section, I investigated the driver of the returns of the MAP portfolios. I used the quadratic regression of Treynor and Mazuy (1966) to test about market timing ability:

$$\text{MAP}_{j,t,L} = \alpha_j + \beta_{j,m} r_{mkt,t} + \beta_{j,m^2} r_{mkt,t}^2 + \epsilon_{jt}, \quad j = 1, \dots, N \quad (8)$$

Where the significantly positive coefficient, β_{j,m^2} , indicates successful market timing ability.

Table 13 presents the result of the market timing regression for Korean, U.S. and Japan markets. In all of the markets, the coefficients are all positive and statistically significant, indicating that successful market timing ability in MA timing strategy. Nevertheless, the alphas of the regression are still positive and significant. The explanatory power of all decile portfolios is also relatively low. This suggests that market timing ability alone does not explain the abnormal returns generated by MA timing strategy in Korean, U.S., and Japan stock markets.

Rank	Korean Stock Market				U.S. Stock Market				Japan Stock Market			
	α_j	$\beta_{j,m}$	$\beta_{j,m}^2$	R^2	α_j	$\beta_{j,m}$	$\beta_{j,m}^2$	R^2	α_j	$\beta_{j,m}$	$\beta_{j,m}^2$	R^2
Low	0.250 (8.47)	-0.032 (-1.48)	2.439 (4.50)	20.46	0.636 (1.87)	-0.0021 (-2.62)	-0.0003 (-0.64)	14.25	0.554 (3.20)	-0.0032 (-3.83)	0.0002 (3.61)	11.64
2	0.344 (12.22)	0.006 (0.43)	3.686 (3.68)	13.06	0.267 (5.17)	-0.0022 (-3.85)	0.0003 (4.16)	30.65	0.379 (3.92)	-0.0037 (-3.43)	0.0002 (3.17)	20.00
3	0.369 (9.94)	-0.058 (-2.19)	4.069 (4.34)	19.02	0.298 (7.36)	-0.0027 (-4.47)	0.0003 (4.07)	56.44	0.764 (2.44)	-0.0046 (-3.99)	0.0002 (3.00)	21.72
4	0.407 (11.03)	-0.092 (-2.45)	4.197 (5.13)	19.67	0.347 (8.21)	-0.0028 (-5.20)	0.0004 (3.97)	48.38	0.597 (3.60)	-0.0043 (-3.99)	0.0001 (1.78)	18.07
5	0.508 (10.09)	-0.095 (-1.95)	4.125 (5.35)	32.24	0.347 (10.28)	-0.0038 (-9.15)	0.0005 (4.89)	52.70	0.366 (8.68)	-0.0048 (-3.61)	0.0003 (3.77)	26.20
6	0.499 (9.61)	-0.081 (-2.07)	4.527 (4.90)	22.77	0.390 (10.91)	-0.0040 (-8.32)	0.0005 (4.69)	56.19	0.440 (8.03)	-0.0047 (-3.60)	0.0003 (4.01)	25.82
7	0.529 (9.77)	-0.088 (-2.00)	5.796 (5.11)	17.17	0.444 (11.01)	-0.0041 (-6.55)	0.0006 (4.64)	50.05	0.393 (7.40)	-0.0049 (-3.56)	0.0003 (2.57)	25.00
8	0.591 (16.51)	-0.057 (-1.42)	6.573 (4.86)	22.56	0.469 (7.38)	-0.0037 (-4.50)	0.0007 (4.55)	30.10	0.524 (11.11)	-0.0046 (-3.55)	0.0004 (3.27)	22.58
9	0.697 (11.77)	-0.052 (-1.25)	6.412 (4.57)	19.69	0.567 (8.94)	-0.0036 (-3.79)	0.0007 (4.76)	27.78	0.535 (6.23)	-0.0051 (-3.55)	0.0003 (2.94)	24.95
High	0.753 (10.61)	0.006 (0.20)	8.904 (4.34)	17.40	0.604 (8.02)	-0.0043 (-4.94)	0.0009 (4.79)	37.88	0.844 (2.72)	-0.0051 (-3.80)	0.0005 (2.91)	23.95
High-Low	0.503 (9.59)	0.038 (1.02)	6.465 (4.06)		-0.0316 (-0.11)	-0.0023 (-1.68)	0.0012 (2.86)		0.290 (1.23)	-0.002 (-2.54)	0.000 (1.24)	

Table 13. Market Timing Regression

6. Conclusion

In this paper, I present supporting evidence that moving average can generate investment profitability when applied to decile volatility portfolios in Korean, U.S., and Japan Stocks markets. The moving average timing strategy generates higher return or outperforms the respective buy-and hold strategy. The moving average timing strategy also performs well in Korean and Japan which have low volatility effect. The low volatility effect disappeared when applied to MA timing strategy. MA timing strategy performance also still generate significant strong result even after considering transaction costs. This positive return of MAP has little risk exposure of CAPM, Fama-French 3 factors model, and also momentum factor. The alpha values for all of the models are significantly positive and have positive relation with volatility decile portfolios. Moreover, the results are robust even when using alternative moving average length and size decile portfolios. The abnormal return of MAPs also cannot be explained by market timing factor although they generate a significant successful market timing strategy.

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한국, 미국, 일본 주식 시장의 개별 주식을 이용한 포트폴리오 투자의 이동 평균 수익성

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요약

본 논문은 한국, 미국 및 일본 주식시장의 십분위(decile) 변동성 포트폴리오를 이용하여, 이동평균 시장예측전략이 매수보유전략에 비교하여 비교우위의 수익성 존재여부에 대한 실증적 연구이다. 검증결과에 의하면, 이동평균 시장예측전략이 매수보유전략에 비교하여 보다 높은 수익을 발생시킨다는 유의적인 결과를 발견하였다. 이 증거는 이동평균 시장예측전략에 의한 매수 및 매도의 거래회수에 대한 거래비용을 고려한 경우에도 여전히 유의적인 수익성을 보여준다. 더욱이 본 연구에서 확인한 증거는 자본자산가격결정모형(CAPM), 3요인모형(Fama & French three-factor model) 등의 위험조정수익률에서도 유의적이다. 이상의 검증결과에서 이동평균 시장예측전략으로부터 확인된 비교우위의 수익성 증거는 초과수익과 가격결정모형에 의하여 설명되지 않는 부분이 존재함에 따라 향후 연구에 대한 새로운 연구주제의 제안이라는 공헌도를 제공한다.